

System Architecture

The proposed system follows a multi-agent architecture in which multiple intelligent agents collaborate to transform an input image into a coherent and creative narrative. The system is designed as a sequential pipeline enhanced with an iterative feedback loop to improve the quality of the generated output.

1. Overview

The architecture consists of four main components: User Interface, Orchestrator, Vision Agent, Narrative Agent, and Critic Agent. Each component is responsible for a specific task within the workflow, ensuring modularity and scalability.

2. User Interface

The user interface is implemented using Streamlit. It allows users to upload an image, select a desired story genre, and view the generated narrative. The interface serves as the entry and exit point of the system.

3. Orchestrator

The orchestrator, implemented using LangChain or LangGraph, manages the interaction between the different agents. It controls the execution flow, ensures proper data transfer, and handles the iterative refinement loop until the output meets the required quality threshold.

4. Vision Agent

The Vision Agent is responsible for extracting semantic information from the input image. It utilizes a pre-trained model such as BLIP to generate a descriptive caption that summarizes the visual content.

5. Narrative Architect Agent

The Narrative Agent takes the generated caption along with the selected genre as input and produces a structured story. This component relies on a Large Language Model (LLM) to generate approximately 200 words of coherent narrative text.

6. Critic Agent

The Critic Agent evaluates the generated story based on multiple criteria, including coherence, grammatical correctness, and consistency of tone. It provides feedback that is used to refine the story.

7. Iterative Feedback Loop

A key feature of the system is the iterative feedback mechanism. If the generated story does not meet predefined quality standards, the Critic Agent sends feedback to the Narrative Agent. The story is then refined and re-evaluated. This loop continues until an acceptable level of quality is achieved.

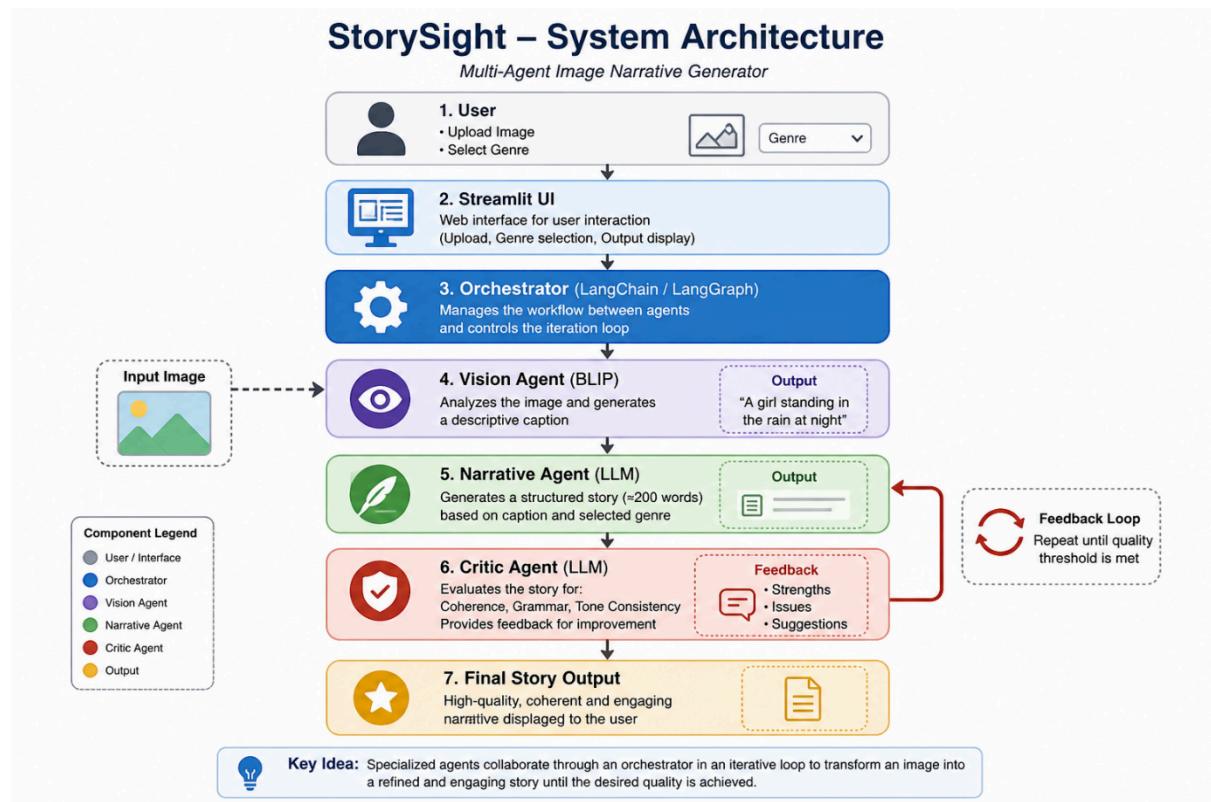
8. Data Flow

The system operates as follows:

1. The user uploads an image through the interface
 2. The Vision Agent generates a caption
 3. The Narrative Agent produces a story
 4. The Critic Agent evaluates the story
 5. Feedback is used to refine the output
 6. The final story is presented to the user
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9. Summary

The architecture adopts a modular and collaborative design, where each agent performs a specialized function. The integration of an iterative feedback loop enhances the overall quality of the generated narratives, making the system more robust and effective compared to traditional single-model approaches.



StorySight system architecture illustrating the multi-agent workflow and the iterative feedback loop used to improve narrative quality.